STATS 200A Discussion 6

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$(\Omega, \mathcal{F}, \mathbb{P}), X, Y : \Omega \to \mathbb{R}$

$$\mathbb{E}(X \mid Y) = \mathbb{E}(X, \sigma(Y)) = Z : \Omega \to \mathbb{R} = \begin{cases} Z \text{ is } \sigma(Y)\text{-measurable} \xleftrightarrow{\text{Doob-Dynkin} Z = g(Y)} \\ Z \in L^1(\Omega) : \int |Z| \, d\mathbb{P} < \infty \\ A \in \sigma(Y) : \mathbb{E}(X \cdot \mathbf{1}_A) = \mathbb{E}(Z \mathbf{1}_A) \end{cases}$$

$B \in \mathcal{F}, B \subseteq \Omega$
$Y := \mathbf{1}_B$
$\omega \in \Omega$
$Y(\omega) = \begin{cases} 1 & \omega \in B \\ 0 & w \notin B \end{cases}$
$\mathbb{E}(X \mid Y) = \mathbb{E}(X \mid \sigma(Y))$
$\sigma(Y) = \{Y^{-1}(B) \mid B \in \mathcal{B}(\mathbb{R})\}$
$Y^{-1}(B) = \{\omega \in \Omega \mid Y(\omega) \in B\}$
$Y^{-1}(\{0\}) = \Omega \backslash B$
$Y^{-1}(\{1\}) = B$
$Y^{-1}(\{0, 1\}) = \Omega$
$Y^{-1}(\mathbb{R} \backslash \{0, 1\}) = \emptyset$
$\sigma(Y) = \{B, B^c, \Omega, \emptyset\}$
$\mathbb{E}(X \mid Y) = \mathbb{E}(X \mid \sigma(Y)) = Z = \mathbb{E}(X \mathbf{1}_B) \cdot \mathbf{1}_B + \mathbb{E}(X \cdot \mathbf{1}_{B^c}) \mathbf{1}_B = \mathbb{E}(X \mathbf{1}_B) \cdot Y + \mathbb{E}(X \mathbf{1}_{B^c})(1 - Y)$
$X \in \{1, \ldots, 6\}$
$Y = 1_{\{X \geq 5\}}$
$\mathbb{E}(X \mid Y) = 5.5 \cdot \mathbf{1}_{X \geq 5} + 2.5 \cdot \mathbf{1}_{\{X < 5\}}$

$$\mathbb{P}(X > 1 \mid Y = 2) \neq \frac{\mathbb{P}(X > 1 \cap Y = 2)}{\mathbb{P}(Y = 2)}$$

Probability Kernal $K(y, B) = \mathbb{P}(X \in B \mid Y = y)$

$$\begin{cases} y \mapsto K(y, B) & \text{measurable } \forall B \in \mathcal{B}(\mathbb{R}) \\ B \mapsto K(y, B) & \text{valid prob measure } \forall y \in \mathbb{R} \end{cases}$$

$$K(y, B) = \mathbb{P}(X \in B \mid Y = y)$$

$$\int x K(y, dx) = \mathbb{E}(X \mid Y = y)$$

$$\int h(x) \, K(y, dx) = \mathbb{E}(h(x) \mid Y = y)$$

$$\int K(y, dx) = \mathbb{E}(1 \mid Y = y) = 1$$

$$\int K(y, B) \, d\mathbb{P}_Y(y) = \mathbb{P}(X \in B)$$

$$\int x^2\, K(y, dx) - \left(\int x\, K(y, dx)\right)^2 = \mathrm{Var}(X \mid Y = y)$$

X, Y jointly continuous

$$K(y, dx) = \frac{f_{X,Y}(x, y)}{f_Y(y)}\, dx$$

If X, Y independent then

$$K(y, B) = \mathbb{P}(X \in B \mid Y = y) = \mathbb{P}(X \in B) = \mathbb{P}_X(B)$$

Example 6.1

$U_1, U_2, U_3 \overset{iid}{\sim} \mathrm{Unif}(0, 1)$

(1) $M := \max(U_1, U_2, U_3)$

(2) $K_1(B, m) = \mathbb{P}(U_1 \in B \mid M = m)$

(3) $\mathbb{E}(U_1 \mid M = m)$

(4) $\mathrm{Var}(U_1 \mid M = m)$

Solution.

(1)

$$\begin{aligned}
\mathbb{P}(M \le m) &= \mathbb{P}(U_1 \cap U_2 \cap U_3 \le m) \\
&\overset{iid}{=} \mathbb{P}(U_1 \le m)\mathbb{P}(U_2 \le m)\mathbb{P}(U_3 \le m) \\
&= m^3 \mathbf{1}_{m \in [0,1]} \\
f_M(m) &= 3m^2 \mathbf{1}_{m \in [0,1]}
\end{aligned}$$

(2)

$$\begin{aligned}
K_1(B, m) &= \mathbb{P}(U_1 \in B \mid M = m) \\
K_1((m, 1], m) &= \mathbb{P}(U_1 \in (m, 1] \mid M = m) = 0 \\
K_1(\{m\}, m) &= \mathbb{P}(U_1 = m \mid M = m) = 1/3 \\
B \subseteq [0, m) \qquad K_1([0, m), m) &= \mathbb{P}(U_1 \in B \mid M = m) = \frac{|B|}{m} \\
K_1(B, m) &= \frac{1}{3}\mathbf{1}_{U_1 \in \{m\}} + \frac{2}{3}\int_{B \cap [0,m)} \frac{1}{m}\, dx = \frac{1}{3}\mathbf{1}_m + \frac{2}{3}\,\mathrm{Unif}(0,)
\end{aligned}$$

(3)

$$\begin{aligned}
\mathbb{E}(U_1 \mid M = m) &= \frac{1}{3}m + \frac{2}{3}\frac{m}{2} \\
&= \frac{2}{3}m
\end{aligned}$$

(4)

$$\begin{aligned}
\mathbb{E}(U_1^2 \mid M = m) &= \frac{1}{3}m^2 + \frac{2}{3}\int_0^m \frac{1}{m}x^2\, dx = \frac{2}{3} \cdot \frac{m^2}{3} \\
&= \frac{5}{9}m^2
\end{aligned}$$

$$(\mathbb{E}(U_1 \mid M = m))^2 = \frac{4}{9}m^2$$
$$\mathrm{Var}(U_1 \mid M = m) = \frac{1}{9}m^2$$

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